

METAL SOLUTIONS

EOS Aluminium Al2139 AM

Material Data Sheet

EOS ALUMINIUM AL2139 AM

Highest Strength Aluminium Alloy in AM

EOS Aluminium Al2139 AM is a 2000-series high strength alloy specifically designed for AM. It demonstrates outstanding performance in elevated temperatures up to 200°C. Excellent strength properties include yield and tensile strength around 500 MPa in heat treated state and allow lightweight part design. The fast and simple heat treatment procedure enables an affordable part production.

MAIN CHARACTERISTICS

- Unmatched strength in temperatures between 50-200 °C
- Robust one-step heat treatment
- Parts can be electropolished and anodized

TYPICAL APPLICATIONS

- Aerospace
- Racing
- Transportation & Mobility
- Lightweight designs

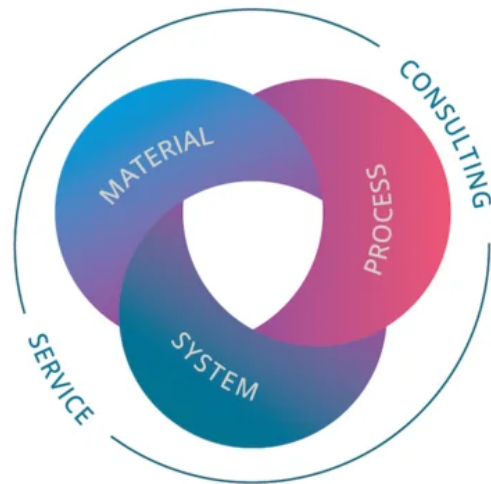
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

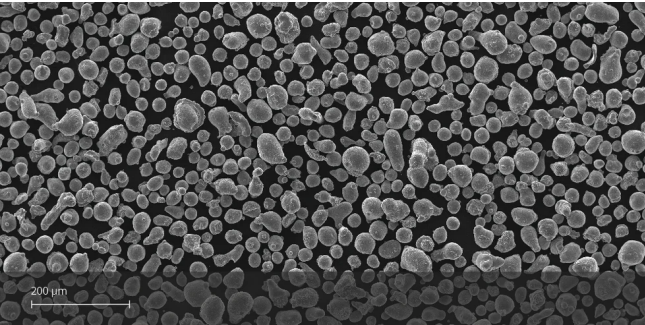
The chemical composition of the EOS Aluminium Al2139 AM powder is modified from Aluminium Association Teal Sheet for Al2139 (wrought aluminium alloys).

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Al		Balance
Cu	4.5	5.5
Mg	0	0.8
Ag	0.15	0.6
Mn	0.2	0.6
Zn	0	0.25
Si	0	0.1
Fe	0	0.15
Cr	0	0.05
V	0	0.05
-	-	4

Powder Particle Size

Generic Particle Size Distribution	20 - 63 µm
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SEM micrograph of EOS Aluminium Al2139 powder

HEAT TREATMENT

Description

EOS Aluminium Al2139 AM recommended T4 Heat Treatment

Solution annealing 45 min @ 490 °C, water quench. Natural aging 3 days in room temperature.

Steps

Step 1: Solution annealing of 45 minutes in 490 °C followed by immediate quenching to water.

Step 2: Natural aging in room temperature for 3 days.

EOS Aluminium Al2139 AM for EOS M 290 | 60 µm

EOS M 290 - 60 µm - TRL 3

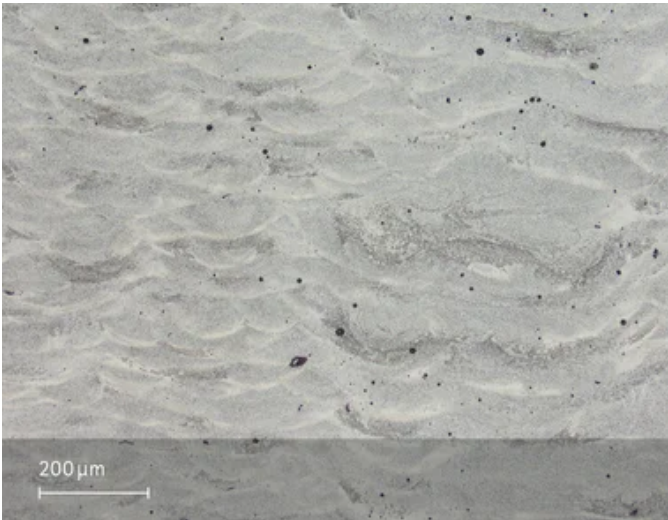
Process Information Metal

Process with good buildability, productivity, and excellent mechanical properties after heat treatment, also at elevated temperatures. Part and support design for high-strength aluminium is required.

System Setup	EOS M 290
EOS Material set	Al2139AM_060_CoreM291_110
Software Requirements	EOSPRINT 2.7 or newer
	EOSYSTEM 2.11 or newer
Recoater Blade	HSS (High Speed Steel)
Build Platform Temperature	175 °C
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	60 µm
Volume Rate	7.2 mm³/s
Min. Wall Thickness	0.4 mm

Chemical and Physical Properties of Parts



Micrograph etched as manufactured

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	60 μm	0.2-0.3 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	60 μm	≥ 2.84 g/cm ³	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 60 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	460	520	4	-	-	-
Horizontal	460	540	6	-	-	-

Mechanical Properties As Manufactured

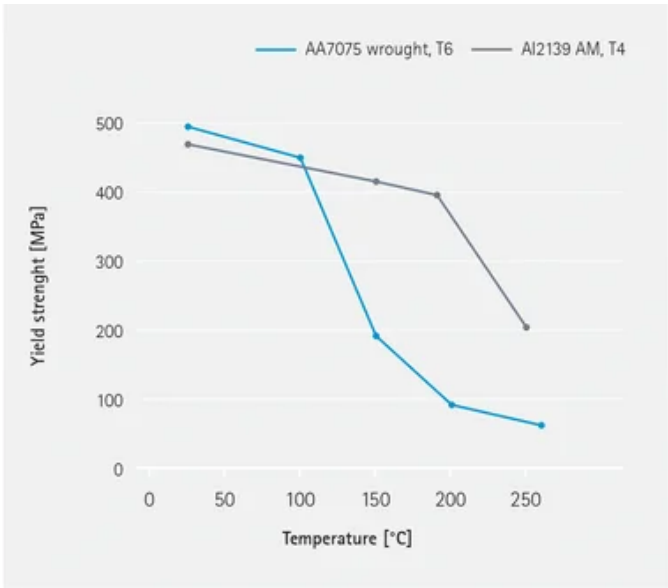
EN ISO 6892-1 Room Temperature 60 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	350	380	6	-	-	-
Horizontal	350	380	8	-	-	-

Hardness

EN ISO 6506-1 Heat Treated	
Value	162
Unit	HBW 2.5/62.5

EN ISO 6506-1 As Manufactured	
Value	112
Unit	HBW 2.5/62.5

Elevated temperature properties



Elevated temperature properties

Elevated temperature tensile testing of EOS Aluminium Al2139 AM (T4) and a wrought grade reference material (T6 HT state) according to EN 2002-002

EOS Aluminium Al2139 AM for EOS M 400-4 | 50 µm

EOS M 400-4 - 50 µm - TRL 3

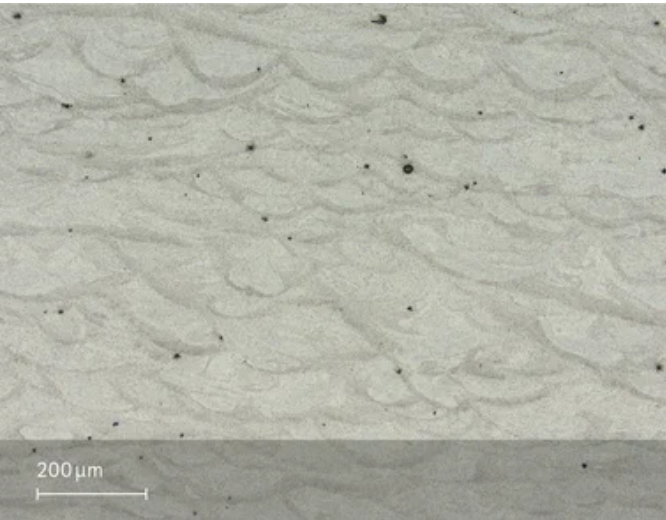
Process Information Metal

High performance process with good microstructure and excellent mechanical properties and low anisotropy after heat treatment. Good mechanical properties at elevated temperatures in heat treated condition. Process offers good buildability of small features with soft recoating. Part and support design for high strength aluminium is required.

System Setup	EOS M 400-4
EOS Material set	Al2139AM_050_CoreM404
Software Requirements	EOSPRINT 2.10 or newer
	EOSYSTEM 2.14 or newer
Recoater Blade	HSS (High Speed Steel)
Build Platform Temperature	195 °C
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	50 µm
Volume Rate	up to 4 x 5.0 mm³/s
Min. Wall Thickness	0.4 mm

Chemical and Physical Properties of Parts



Micrograph etched as manufactured

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	50 μm	0.1 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	50 μm	≥ 2.84 g/cm ³	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 50 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	450	530	6	-	-	-
Horizontal	460	550	7	-	-	-

Mechanical Properties As Manufactured

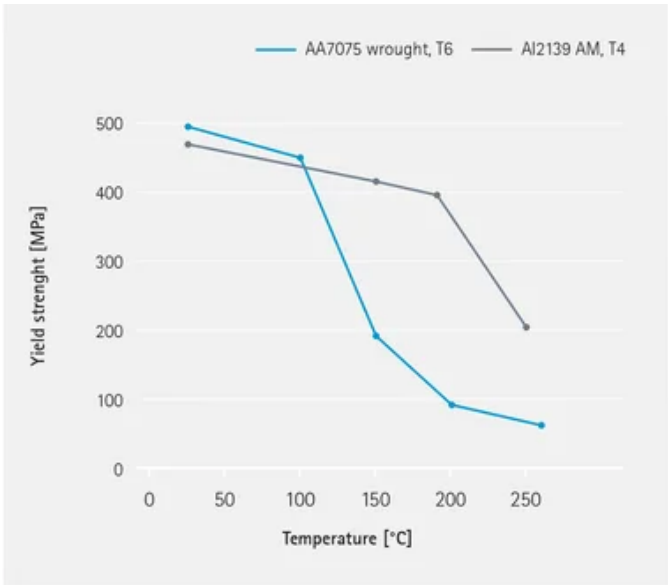
EN ISO 6892-1 Room Temperature 50 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	380	430	8	-	-	-
Horizontal	380	430	10	-	-	-

Hardness

EN ISO 6506-1 Heat Treated	
Value	168
Unit	HBW 2.5/62.5

EN ISO 6506-1 As Manufactured	
Value	115
Unit	HBW 2.5/62.5

Elevated temperature properties



Elevated temperature properties

Elevated temperature tensile testing of EOS Aluminium Al2139 AM (T4) and a wrought grade reference material (T6 HT state) according to EN 2002-002.

HEADQUARTERS

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This powder has not been developed, tested or certified as a medical device according to Directive 93/42/EEC (MDD) or Regulation (EU) 2017/745 (MDR) and is not intended to be used as a medical device, in particular for the purposes specified in Art. 2 No. 1 MDR. Insofar as you intend to use the powder as raw material for the manufacture of pharmaceutical products or medical devices (e.g. as raw material which as a material must meet the requirements of Annex 1, Chapter II MDR), the responsibility and liability for all analyses, tests, evaluations, procedures, risk assessments, conformity assessments, approval and certification procedures as well as for all other official and regulatory measures required for this purpose shall lie solely with you both with regard to the pharmaceutical product and/or medical device manufactured by you and with regard to the properties, suitability, testing, evaluation, risk assessment, other requirements for use of the powder as raw material. In this respect, the limitations of liability pursuant to our General Terms and Conditions and the system sales or material contracts shall apply.

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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